such situations either. The early universe was fabulously small, suggesting that quantum physics was needed, but also fabulously dense, requiring the forbidding mathematical machinery of general relativity. Unfortunately, a theory unifying general relativity with quantum physics hadn't been found, despite having been sought for decades by an army of physicists, including Einstein. As late as the 1960s, some people had suggested that no such unification was necessary: Léon Rosenfeld had said (like a good positivist) that because quantum gravitational effects could never be seen, there was no need to develop a theory to address such unobservable phenomena. But, as general relativity became more reputable, the need to unify it with quantum field theory had become more pressing. By the 1990s, ideas like Rosenfeld's were as far outside the mainstream of physics research as the old claims that cosmology shouldn't be taken seriously. A theory of quantum gravity—now nicknamed a "theory of everything"—was widely considered to be the single most important unsolved problem in all of physics. The most promising candidate was string theory, whose abstruse mathematics yielded glimpses of elegant connections between quantum physics and general relativity. By the early 2000s, combining string theory and inflation seemed like the best hope for a theory of the early universe.

Surprisingly, both string theory and inflation, which were developed quite independently, seem to point to a common conclusion: the existence of a multiverse, an enormous number of multiple independent universes. According to inflation, the universe is unable to escape "eternal inflation": as inflation ends in one part of the universe, it continues in others, and "bubbles" of noninflating universe continually appear in the inflating region. We live in one of these bubbles; other bubbles would be their own universes, cut off from all the others, and each might have its own laws of physics and assortment of fundamental particles. And because inflation is eternal, there would be an infinity of these bubbles—an infinite multiverse of inflation. String theory, meanwhile, doesn't describe a single universe but instead describes a "string landscape," a phenomenally huge number of possible universes—10^{500} or more.

The similarities to the many-worlds interpretation's multiverse were not lost on quantum cosmologists. The appearance of multiverses independent of Everett's interpretation made its strange profusion of worlds downright appealing. Some physicists even proposed that all three of these multiverses